

Enterprise Autonomous Recruitment Operating System

An AI operating system for talent acquisition. Auditable, governed, and built to separate intelligence from execution.

"Where a traditional ATS asks 'where is this candidate?', EAROS asks 'what should we do next?' — and shows the reasoning, one policy-gated step at a time."

What EAROS is

EAROS is an AI operating system for talent acquisition. It separates *intelligence* (Planner, Policy Engine, Reflection) from *execution* (deterministic capabilities in a Registry), so every AI decision is auditable, replaceable, and governed by explicit policy — not buried inside an LLM prompt.

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Part I · Understanding the platform

1. What is EAROS?

EAROS stands for *Enterprise Autonomous Recruitment Operating System*. It is an executive-class demonstration of what happens when you stop bolting AI onto a legacy ATS and instead build a **runtime** for AI recruitment work — one where every action is planned, policy-checked, executed, and reflected on.

Where a traditional ATS asks "*where is this candidate?*", EAROS asks "*what should we do next?*" and shows the reasoning.

Three things that make EAROS different

	Traditional ATS + AI plugin	EAROS
Intelligence	LLM decides <i>and</i> acts in one prompt	Planner (LLM) proposes a plan; Runtime (deterministic) executes it
Governance	Audit trail is best-effort text logs	Every step emits an immutable, replayable event
Risk	AI can create offers no human signed off on	Sensitivity + confidence trigger explicit human approval gates

Who it is for

- **Chief People Officers** who need to prove to a board that AI recruitment is safe
- **VPs of Talent Acquisition** who need to scale a team from 3 to 30 recruiters without losing hiring bar
- **Recruiting Operations** leaders who need an auditable system-of-record for AI decisions
- **CIOs / CTOs** evaluating agentic architectures before rolling out enterprise-wide

2. Core capabilities

EAROS ships with **five architectural capabilities** that are exposed to end-users through a set of purpose-built workspaces:

2.1 Planner (LLM-backed)

Takes a natural-language goal and produces a schema-validated multi-step plan. Every step declares: - The capability it calls (e.g. `cap.screen_candidate`) - Its inputs (typed) - The AI's confidence (0–1) - The sensitivity level (Public / Internal / Confidential / Restricted)

The Planner is a **proposer, not a doer** — it never touches the world state directly.

2.2 Runtime (deterministic executor)

Executes the Planner's steps in order. For each step it: - Retrieves the capability from the Registry - Passes it through the Policy Engine - Runs it and records the outcome as an immutable event - Advances or halts based on the Governance response

2.3 Policy Engine (rule-based)

Evaluates every step against declared policies: - **Fairness** — no protected-attribute inputs to ranking - **PII** — sensitive fields never leave the workspace boundary - **DEI** — diverse-slate enforcement on shortlists - **Compensation** — offer bands + level guardrails - **Sensitivity** — CONFIDENTIAL steps require human approval

Result is one of: `approved`, `blocked`, `requires_approval`.

2.4 Capability Registry

The catalog of every callable action: - `cap.source_candidates` · parallel scan across 10 channels - `cap.parse_resume` · structured extraction - `cap.draft_outreach` · per-tone message generation - `cap.screen_candidate` · 8-dimension rubric - `cap.schedule_interview` · calendar coordination - `cap.generate_offer` · comp-aligned offer letter (always CONFIDENTIAL) - ...and more

Capabilities are **versioned, swappable, and independently testable**. You can replace `cap.screen_candidate` v1 with v2 without redeploying the Planner.

2.5 Governance & Reflection

- **Immutable event log** — every plan, invocation, policy decision, and world-state change is appended and correlation-id-linked
- **Reflection Agent** — after every meaningful execution, writes a report of what worked, what failed, and what to adjust next time — informing future plans

3. Features by user persona

3.1 For Recruiters (`/recruiter`)

- **Autonomy toggle** — flip between *Manual*, *Semi-auto*, and *Full-auto* per requisition
- **Pipeline Intelligence strip** — live per-req stats (active count, avg fit, risk flags, offers)
- **Ops Toolbar** — email/SMS campaigns, screening blasts, resume harvester — all with recipient checklists and paced sends
- **Stage transitions** — manually advance candidates with an audit event fired on every move
- **Simulate Plan** — see the Planner's proposed sequence before it runs

3.2 For Hiring Managers (`/dashboard`, `/intake`)

- **Clickable KPI dashboard** — every tile drills into its underlying records
- **Job Architecture Agent** — paste a brief in English, get a JD + sourcing plan + a matched requisition in seconds

- **What-if hiring simulations** — see how a delay or headcount change moves the trajectory

3.3 For Executives (`/executive`)

- **Workforce simulation** — attrition, budget, expansion, and AI-headcount levers with per-month trajectory
- **AI Recommendation card** with reasoning, evidence, tradeoffs, and residual risks
- **Skill demand + department health** — where the org is bottlenecked and by how much

3.4 For Candidates (`/candidate`)

- **Status lookup** — where they are in the process and what's next
- **Interview prep** — 5-stage walkthrough with STAR-format tips
- **Org overview** — company snapshot, values, departments
- **Day in the life** — realistic hour-by-hour schedule

3.5 For Admins / Auditors (`/governance` , `/agents` , `/deep-dive`)

- **Immutable audit log** with correlation-id search
- **Agent registry** — health, latency, cost, recent activity per agent
- **Deep-Dive replay** — swimlane view of every agent-to-agent event in an execution
- **Layered architecture diagram** — click any node for its role

4. The architecture in one page

EAROS is organised in **five layers**. Every feature you see in the UI is one of these layers rendered in a different way.

```
BUSINESS LAYER · the "why"
Policy Manager · Objectives · Governance

DECISION LAYER · the "what"
Planner · Policy Engine · Strategy Agent · Reflection

EXECUTION LAYER · the "how"
Runtime · Capability Registry · Sourcing · Outreach · Screen

KNOWLEDGE LAYER · grounded facts
World State · Episodic Memory · Skill Graph

INFRASTRUCTURE LAYER · plumbing
Event Bus · Auth (Google OAuth) · Observability · MongoDB
```

Dataflow: *Business steers → Decision proposes → Execution acts → Knowledge grounds → Infrastructure logs everything.*

You can see this diagram live at [/deep-dive](#) → click **ARCHITECTURE** in the top-right.

5. How the AI is governed

Governance in EAROS is not a report — it is a runtime.

Every AI action generates an event

Events are appended to an immutable log with a correlation id that links them into an *execution* you can replay. Event types include:

- `PLAN_CREATED` · Planner proposed a plan
- `POLICY_EVALUATED` · Policy Engine allowed or blocked a step
- `CAPABILITY_INVOKED` · Runtime called a capability
- `APPROVAL_REQUESTED` · a CONFIDENTIAL step needs a human
- `APPROVAL_DECIDED` · a human approved / rejected
- `CANDIDATE_STAGE_CHANGED` · pipeline moved
- `REFLECTION_RECORDED` · what was learned

Human approval gates are real

Steps marked `sensitivity=CONFIDENTIAL` (offers, comp negotiations, terminations, PII exports) always require an explicit human click. There is no auto-approve. If nobody clicks within 10 minutes, the step is **cancelled** — not silently released.

Every recommendation is explainable

The Planner outputs its full reasoning chain: goal parsing → world-state retrieval → capability matching → step scoring. You can see this live at `/planner`.

Part II · Using the platform

6. Getting in

Preview (dev): `https://hiring-runtime.preview.emergentagent.com` **Production:** `https://hiring-runtime.emergent.host`

Two ways to sign in:

Method	When to use
Continue with Google (Emergent-managed OAuth)	Real users on production. Role inferred from email.
Demo buttons (Recruiter · Hiring Manager · Executive · Candidate · Admin)	Instant, no email. Ideal for live demos.

Tip · The Executive demo unlocks the workforce-simulation What-If sliders. The Recruiter demo unlocks the operational studios. Start with **Recruiter** for most demos.

7. Mission Control

Route: `/mission`

The persistent top-nav shows the six primary surfaces. The main page shows:

- Top P0 requisitions and aging offers
- A live event ticker (last N events through the runtime)
- Quick-start scenario launchers

Every demo starts here — this is your context-setting screen.

8. Demo Scenarios — the 15-minute hero flow

Route: `/scenarios`

The **centerpiece** of any executive walkthrough. Each scenario runs the complete lifecycle — Intake → JD → Sourcing → Screening → Outreach → Interview → **Offer (held for human approval)** → Reflection — with pacing so viewers can watch each agent think.

How to run one

1. Pick a scenario card (e.g. *Salesforce Architect · Austin*)
2. Click **RUN SCENARIO**
3. Watch the live panel: - Current-step banner naming the active agent - Progress bar with step counter (*Step 6 / 14 · 43%*) - Per-step outputs as they complete
4. At step 13 the scenario **pauses** with an amber card:

HUMAN APPROVAL REQUIRED Offer for Shreya Chatterjee Policy requires human approval for CONFIDENTIAL offers above L2. AI confidence 0.86. **SCENARIO CANCELS IN** 9:47

[APPROVE OFFER] [REJECT]

1. Click **APPROVE OFFER** to release the gate — Reflection runs and the scenario completes
2. Use **OPEN REPLAY** → to see the swimlane replay, or **AUDIT** → to see every recorded event

What this proves · Policy actually gates AI actions. The pause is real — you can wait 10 minutes; nothing auto-approves.

9. Hiring Intake → Job Architecture Agent

Route: `/intake`

Paste a plain-English brief (or use a pre-loaded example). The **Job Architecture Agent** flow lights up in four steps:

1. **Requirements Agent** — extract must-haves, seniority, location, comp, urgency
2. **Job Architecture Agent** — generate a calibrated JD with responsibilities + comp band
3. **Sourcing Strategy Agent** — 3-wave sourcing plan with Boolean search string
4. **Recruiter Handoff** — match to the closest existing requisition

Click **OPEN IN RECRUITER COPILOT** → and the matched req is pre-selected.

Resilience · If the LLM stalls, the agent retries 3× on the frontend and 2× on the backend, then falls back to a deterministic parser. Users never see a raw gateway error.

10. Hiring Dashboard — drill-downs

Route: `/dashboard`

Six KPI tiles — each is **clickable** and opens a right-side drill drawer:

KPI	Drill-down shows
Open reqs	Active requisitions with priority + location + days-open
In pipeline	Every non-terminal candidate
Offers active	Everyone in offer stage
Hired YTD	Everyone marked hired
Avg time-to-fill	Reqs sorted worst-first
P0 roles	Board-level critical priorities

Every drill-down row deep-links into the Recruiter Copilot with the correct requisition pre-selected.

11. Recruiter Copilot — the operator surface

Route: `/recruiter` (or `/recruiter?job=<job_id>` to preselect)

Autonomy toggle

Flip between three modes at any time:

- **Manual** — AI recommends. Every action needs a click.
- **Semi-auto** — AI acts on low-risk steps; humans approve offers, comp, comms.
- **Full-auto** — AI runs the full lifecycle within policy. Humans audit after.

Pipeline Intel strip

Live per-req metrics: Active · Avg fit % · Risk flags · Offers · Autonomy mode.

Ops toolbar

Tool	What it does
Email campaign	Multi-select recipients → subject + body with <code>{{first_name}}</code> / <code>{{role}}</code> templating → paced send with audit trail
SMS campaign	Short-form outreach with the same recipient checklist
Screening blast	Fires async AI screening to N candidates in parallel
Resume harvester	Drop <code>.pdf/.docx</code> — auto-parse, dedupe against ATS

Candidate detail

Selecting a candidate shows their profile, fit score, skill chips, and a **stage transition** dropdown. Every transition emits an immutable `CANDIDATE_STAGE_CHANGED` audit event.

Simulate plan

Click **SIMULATE PLAN** to see the Planner build a multi-step plan for the selected candidate.

12. Sourcing — parallel sweep + action plan

Route: `/sourcing`

1. Pick a requisition
2. Click **RUN SWEEP** — 10 channels scan in parallel: GitHub · LinkedIn · Dice · Internal ATS · Referrals · Naukri · Hired · Stack Overflow · AngelList · Gem. Each returns match counts + quality scores.
3. Summary cards: Total matches · Unique candidates · Duplicates removed · Weighted quality
4. **Recommended Action Plan** appears with 4 executable next steps: - `cap.draft_outreach` — Contact top 25 (→ Outreach Studio) - `cap.screen_candidate` — AI screen top 10 (→ Screening) - `cap.retarget` — Retarget silver-medalists from the top source - `cap.schedule_interview` — Schedule 5 with the hiring manager

Each button turns green after you trigger it.

13. Resume Studio

Route: `/resume`

Paste a resume + pick a target job. The parser extracts skills, experience, comp expectations, and right-to-work signals. Then pick a **format**:

- **One-page summary** — compact recruiter-facing card
- **Client submission** — full anonymised package with skills matrix (verified vs role), differentiators, comp/logistics footer, PII-redacted watermark

Both formats have download buttons.

14. Outreach Studio

Route: `/outreach`

Left column — searchable multi-select **recipient list**. Filter by name / title / location.

Right column — pick a template-source candidate → click **GENERATE PACK**. The AI produces:

- **3 email variants** — Warm / Neutral / Formal — each with response probability, editable subject + body, and a **SEND TO N** button
- **LinkedIn InMail · SMS · WhatsApp · Voice Script** — each with a send button

Sending triggers a **paced send** with a live "Sending..." → "Sent · tracked in Governance" banner.

15. AI Interview Suite

Route: `/interview`

One suite, **four modes**:

Mode	Use case	Report contains
AI Screening	5-minute async structured screen	8-dimension rubric with scores, weights, notes; composite + band
AI Voice	Live 12-minute conversational phone screen	Transcript signals (Technical Depth, Structured Thinking, Motivation, Culture, Red Flags), followups
AI Copilot	Live human interview with real-time AI assist	Live per-area scoring, AI-suggested followups, AI flags, scorecard draft
Autonomous Video	Fully AI-driven async video	Per-question scores, integrity (face-match, screen capture), delivery (confidence, WPM, filler words)

Pick a mode → candidate → optional context → action button. A **PhaseTicker** (Warming → Listening → Scoring) plays before the report renders.

16. Executive Copilot — What-If simulations

Route: **/executive**

Five KPI cards on top (headcount, attrition YTD, open reqs, offers active, WFP alignment).

What-If sliders

- **Attrition %** · dial current 12% up to 35%
- **Budget delta %** · $\pm$ vs baseline
- **Hiring freeze** · checkbox
- **Bangalore expansion** · checkbox
- **AI Engineering doubles** · checkbox
- **Horizon** · 6 / 12 / 24 months

Click **RUN SIMULATION** — you get a starting → ending headcount, a per-month trajectory chart, and an AI Recommendation card with **reasoning** · **evidence** · **tradeoffs** · **residual risks**.

Below

- **Department health** — capacity, hiring velocity, at-risk teams
- **Skill demand** — top skill gaps by scarcity × business criticality

17. Candidate Assistant

Route: `/candidate`

The candidate-safe view. Four tabs:

- **Status** — enter a candidate id → stage + next step
- **Interview prep** — 5-stage walkthrough with STAR tips
- **Org overview** — company snapshot, values, departments
- **Day in the life** — hour-by-hour schedule

No confidential data ever crosses this boundary.

18. Intelligence Tab (Planner · Agents · Deep-Dive)

Planner — `/planner`

Type a goal + optional JSON context → **GENERATE**. The **Planner Reasoning Trace** shows 5 phases ticking through:

1. Parsing goal
2. Retrieving world state
3. Matching capabilities
4. Scoring & ordering steps
5. Schema-validating plan

Dispatch footer shows step count · unique capabilities · avg confidence.

Agents — `/agents`

Each agent has a card with: name · version · health · 24h runs · latency · cost · inputs · outputs · capability dependencies.

Click any card to expand: - **Recent activity** · **last 24h** — 5 timestamped events (ok / held) - **Current / queued tasks** — 3 tasks with priority + ETA

Deep-Dive — `/deep-dive`

Two views (toggle in the top right):

- **SWIMLANE** — pick a past execution → replay every agent-to-agent event with real payloads, confidence, sensitivity, and policy hits
- **ARCHITECTURE** — the 5-layer interactive diagram. Click any node for its role.

19. World State & Governance

World State — [/world](#)

Seven sub-tabs, all switch live:

- **Organization** — LevelShift snapshot
- **Departments** — 5 departments with size + focus
- **Teams** — 9 teams with charter + capacity
- **Jobs** — every req with priority + location + status
- **Candidates** — all 152 candidates with stage + fit score
- **Offers** — offers in flight
- **Skills** — the skill graph with supply/demand

Governance — [/governance](#)

The immutable audit log. Filter by actor, subject, event type, or time window. Every row is signed with a correlation id — click to jump to the swimlane replay.

Part III · Playing it well

20. The recommended 15-minute executive demo script

Copy-paste this. Timing is deliberate.

Time	Screen	What to show
0:00	Landing	Click Recruiter demo — land on Mission Control
0:30	<code>/scenarios</code>	Run <i>Salesforce Architect · Austin</i> . Narrate: "Watch each agent think — it's paced so you see the reasoning."
3:00	<code>/scenarios</code>	The scenario hits the Human Approval Gate — pause here. "Policy requires a human. AI doesn't cross this line." Click APPROVE .
4:00	<code>/scenarios</code> → OPEN REPLAY	Show the swimlane replay. "Every event is immutable."
5:30	<code>/intake</code>	Paste a fresh brief. "Job Architecture Agent — 4 sub-agents, 6 seconds." Click handoff.
7:00	<code>/recruiter?job=...</code>	Flip to Full-auto . Show the ops toolbar. Launch an Email campaign to 3 candidates.
9:00	<code>/executive</code>	Slide attrition to 22%. "That's a \$4M scenario." Click RUN SIMULATION .
11:00	<code>/deep-dive</code> → ARCHITECTURE	Show the 5-layer diagram. "Intelligence separated from execution."
13:00	<code>/governance</code>	Filter to <code>POLICY_EVALUATED</code> . "Every AI decision is auditable."
14:30	Q&A	

Talking points that land

- "The AI never touches the world state directly. It proposes; the Runtime disposes."
- "CONFIDENTIAL is not a label — it's a runtime gate."
- "Reflection means the system gets better between demos, not just between releases."

- "Every capability is versioned. We can swap the screener from v1 to v2 without redeploying the Planner."
-

21. FAQ & Troubleshooting

Q: The scenario reached the approval gate but nothing happens when I click APPROVE. Refresh and re-run. Rarely, the polling loop can lose the execution id after a very long idle period. Server state is preserved for 24 hours.

Q: The intake flow shows a "TRY AGAIN" button. The upstream LLM was briefly unreachable. The backend has already fallen back to a deterministic parser that produces a credible intake. Click **TRY AGAIN** or the **Analyze** button — you will get a valid result within ~50 seconds worst case.

Q: I'm on the executive demo but I don't see the What-If sliders. Log out (top-right) and click the **Executive** demo button on the landing page — not Recruiter.

Q: How do I reset the demo data? The seed re-runs on every backend restart. All 152 candidates, 12 requisitions, and 5 scenarios are idempotent. Contact Emergent Support to restart the pod.

Q: Where do I download the report / signed offer? Every generated artifact has a download affordance in the studio that produced it (Resume Studio, Outreach Studio, Interview Suite reports). Storage is Governance-tracked — nothing leaves the system unlogged.

Q: Can I try Google OAuth instead of the demo buttons? Yes — click **Continue with Google** on the landing page. Your role is inferred from the email domain.

Q: Preview vs Production — where should I demo? Demo from **Production** (<https://hiring-runtime.emergent.host>) — it's the version your buyers will use. Preview is for testing changes before deploy.

22. Glossary

Term	Meaning
Capability	A callable, versioned action (e.g. <code>cap.screen_candidate</code>) registered in the Capability Registry
Confidence	The Planner's self-assessed likelihood the step will succeed (0–1)
Correlation ID	A single id shared by every event in one logical execution
Execution	A run of a plan — one goal, N steps, one correlation id
Governance	The immutable audit log + approval workflow
Planner	The LLM-backed proposer; produces schema-validated plans
Policy Engine	The rule-based evaluator; approves, blocks, or requires-approval
Reflection	Post-execution report: what worked, what failed, what to change
Runtime	The deterministic executor of Planner steps
Sensitivity	One of Public / Internal / Confidential / Restricted — drives approval requirements
World State	The grounded facts (jobs, candidates, offers, skills) — the ATS surface

Support

- **Preview & bugs** · your Emergent success manager
- **Production incidents** · Emergent Support
- **Feature requests / feedback** · commits with `feat:` prefix are tracked

EAROS · Planner → Runtime → Policy Engine → Capability Registry → Governance. Every intelligence decision is separated from execution — auditable, replaceable, and human-gated where it matters.

EAROS · PLANNER → RUNTIME → POLICY → CAPABILITY → GOVERNANCE